

Transient Analysis of Non-uniform Windings or Lines with Non-linear Termination

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Abstract

The paper deals with the transient analysis of non-uniform transmission lines and machine windings with non-linear terminations. The proposed time-domain technique is characterized by a relatively small computational burden. The analytical solution of the derived system of simultaneous differential and algebraic equations is done in Mathematica. The procedure is applied to several representative case studies involving line and winding non-uniformity and/or the non-linearity of the load termination, such as diodes and saturated coils. Step function and 50-Hz sinusoidal voltage sources are considered. The results include, among others, the time response of the source and load currents, as well as the current and voltage distributions.

Keywords: Distributed parameter circuits, equivalent circuit, inductances, lines, machine, modeling, mutual non-uniform non-linear loads, termination, diodes, transients, winding

INTRODUCTION

Most of the documented investigations on the transient analyzes of lines and machine windings are addressing cases dealing with uniformly-distributed linear circuit parameters and linear-load terminations [1–9]. A small number of these studies deal with non-uniform parameter distributions such as those involving line towers [2] or river crossings [3–6, 8]. A much smaller number of these cases could address both non-uniform and non-linear situations such as the presence of power electronic devices, surge arresters, corona discharges and iron-cored coils [7, 8]. The main objective of this paper is to investigate these cases based on a time-domain analysis. The first step is the proper sectionalization of the line or machine winding. The two Kirchhoff's circuit laws will then be applied in order to write a system of simultaneous differential and algebraic equations. Their solution yields all the unknown currents and voltages. As representative examples, the transients in this paper are initiated by a voltage source of a step function or sinusoidal time waveform. The two non-linear cases of an ideal diode and an iron-cored coil will be investigated.

The results pertinent to several case studies will be presented and discussed.

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THE METHOD OF ANALYSIS

As an illustrative example, Figure 1 illustrates a five-section equivalent circuit of a transformer winding. Each section is represented by a series resistance r , a shunt ground capacitance c_g and a series capacitance c_s . The sections exhibit the self- and mutual inductances explained below. The number of sections will determine how many resonance frequencies will appear in the simulation results. Increasing this number will lead to a proportional rise in the number of state variables (the inductor currents and capacitor voltages).

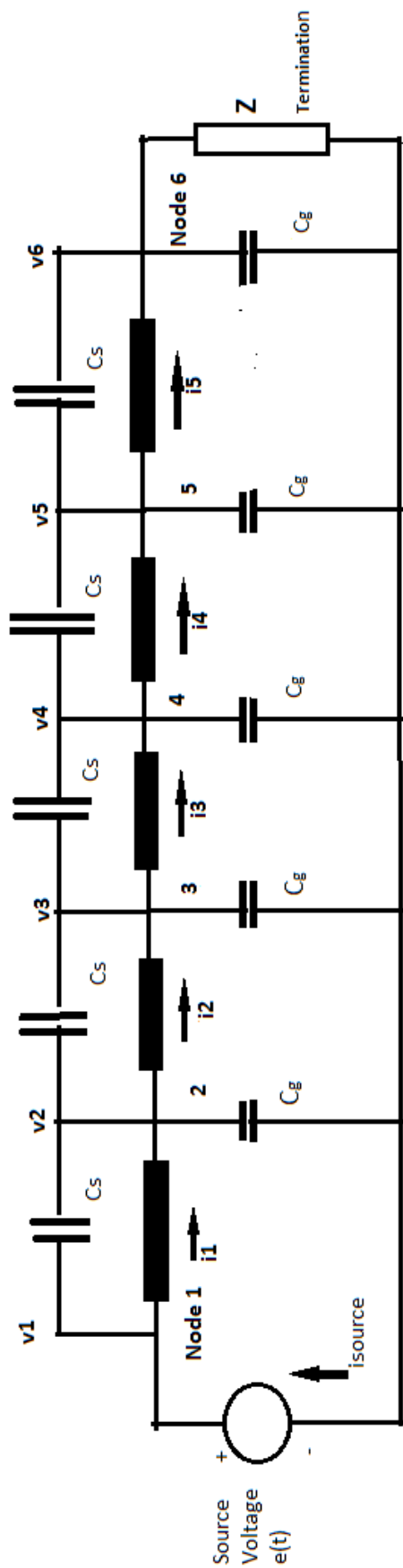


Figure 1. The assumed equivalent circuit.

The Circuit Parameters and the System Equations

- The self-inductance per section $m_0=29 \times 10^{-3}$ H;
 - The mutual inductance between two winding sections separated by one section $m_1=13.573 \times 10^{-3}$ H;
 - The mutual inductance between two winding sections separated by two sections $m_2=6.231 \times 10^{-3}$ H;
 - The mutual inductance between two winding sections separated by three sections $m_3=3.379 \times 10^{-3}$ H;
 - The mutual inductance between two winding sections separated by four sections $m_4=1.987 \times 10^{-3}$ H;
- $cg=8.5 \times 10^{-8}$ F;
 $r=10$; (The Resistance per Section)
 $cs=3.4 \times 10^{-12}$ F; (The Series Capacitance per Section)
 $tend=0.03$; (The Time Range)
The unit-step source volte e is constant and thus has a zero-time derivative.
 $u[x_]=1/2+(1/\pi) \times \text{ArcTan}[1500 x]$; (an approximate definition of the unit-step function)

The Solution of the Simultaneous Equations

It should be noted that the current and voltage first derivatives in the following equations are with respect to the time t .

$$\text{sol}=\text{NDSolve}[\{e-v_2=i_1 r+i_1' m_0+i_2' m_1+i_3' m_2+i_4' m_3+i_5' m_4 \quad (1)$$

$$v_2-v_3=i_2 r+i_1' m_1+i_2' m_0+i_3' m_1+i_4' m_2+i_5' m_3 \quad (2)$$

$$v_3-v_4=i_3 r+i_1' m_2+i_2' m_1+i_3' m_0+i_4' m_1+i_5' m_2 \quad (3)$$

$$v_4-v_5=i_4 r+i_1' m_3+i_2' m_2+i_3' m_1+i_4' m_0+i_5' m_1 \quad (4)$$

$$v_5-v_6=i_5 r+i_1' m_4+i_2' m_3+i_3' m_2+i_4' m_1+i_5' m_0 \quad (5)$$

$$i_{\text{source}}=i_1+cs \times (e'-v_2') \quad (6)$$

$$i_1=i_2+cg v_2'+cs (v_2'-v_3')+cs (v_2'-e') \quad (7)$$

$$i_2=i_3+cg v_3'+cs (v_3'-v_4')+cs (v_3'-v_2') \quad (8)$$

$$i_3=i_4+cg v_4'+cs (v_4'-v_5')+cs (v_4'-v_3') \quad (9)$$

$$i_4=i_5+cg v_5'+cs (v_5'-v_6')+cs (v_5'-v_4') \quad (10)$$

$$i_5=cg v_5'+cs (v_6'-v_5')+i_{\text{load}} \quad (11)$$

$$v_6=i_{\text{load}} \times (rf \times u[i_{\text{load}}]+rr \times u[-i_{\text{load}}]) \quad (12)$$

The current i_{load} is the current through the termination impedance Z .

The above 12 simultaneous equations can be analytically solved through the application of the command (NDSolve) within the software *Mathematica* (Versions 7.0, 10.0 or 12.2). The program output will be 12 interpolating functions, which are closed-form time-domain expressions of the unknown voltages and currents. They are the seven currents ($i_1, i_2, i_3, i_4, i_5, i_{\text{source}}$ and i_{load}) as well as the five voltages v_2, v_3, v_4, v_5 , and v_6 . These expressions will be functions of the winding circuit parameters and the termination impedance Z . For more details, reference [8] should be consulted. It should be noted that the expression describing the voltage source e can handle any time waveform. For the sake of demonstration, this paper will deal with step-functions and sinusoidal source voltage waveforms.

As indicated in equation (12), the ideal diode will be simulated using the idealized voltage/current characteristics of the shape depicted in Figure 2. It is expressed by the following equation in terms of the unit-step function $u(x)$:

$$v=i(R_f.u(i) +R_r.u(-i)] \quad (13)$$

where R_f and R_r are the diode's resistances in the forward and reverse directions, respectively. This representation was first introduced by Saied in [9].

SAMPLE RESULTS

The Response of a Non-uniform Winding Excited by a 100-V Step Voltage Source and Terminated by a Diode

The plot in Figure 3(a) illustrates the source current injected into the non-uniform line when excited by a 100 V step voltage source and loaded by a diode of $R_f=0$ and $R_r=1M\Omega$ over the time range $0 \leq t \leq 30ms$. The transient source current can momentarily reach 0.2A. It includes several high frequency components having a dominating frequency close to 700Hz. They are almost completely damped out after about 30ms. Figure 3(b) shows the time response of the current i_3 flowing through the middle section of the winding. It is similar to the source current but has smaller magnitudes, as expected. The voltage v_3 at the junction between the second and third sections is given in the plot in Figure 3(c). It assumes the maximum value 185V and approaches its expected final magnitude 100V after 30ms as well. It exhibits a small negative value immediately after switching the voltage source at $t=0$. The plot of the load voltage, v_6 , is illustrated by in Figure 3(d). It reaches maximum positive and negative values of about +215V and -115V, respectively. Its final value is again 100V.

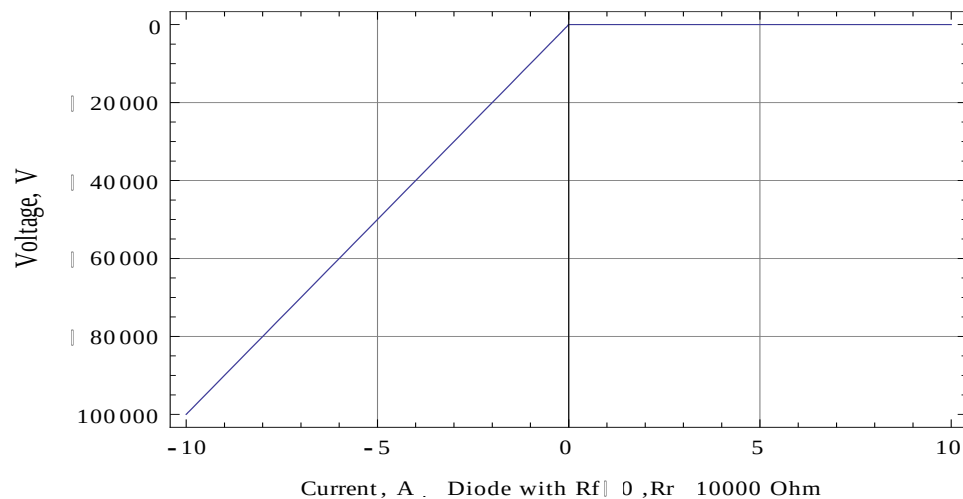
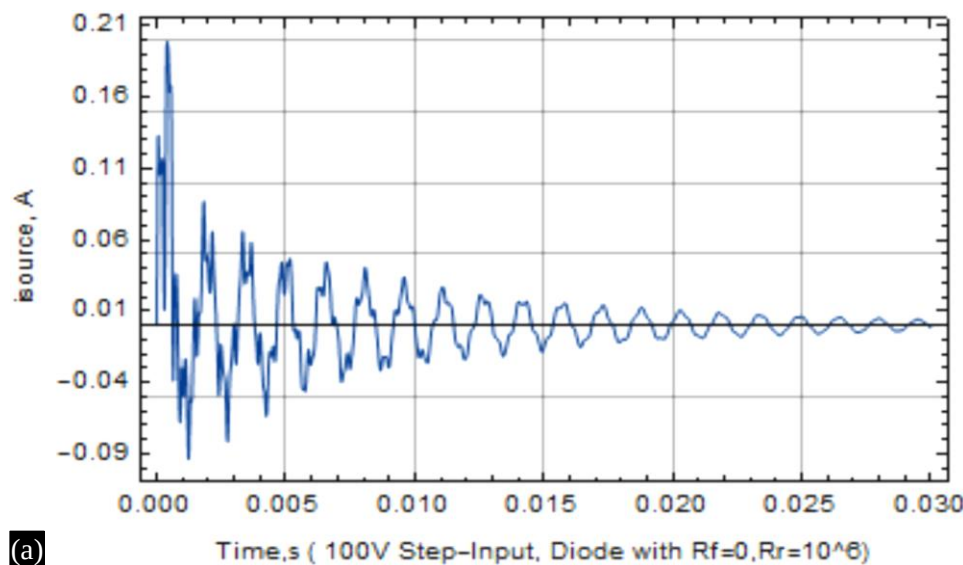


Figure 2. The assumed idealized diode's voltage-current characteristic, according to reference [9].



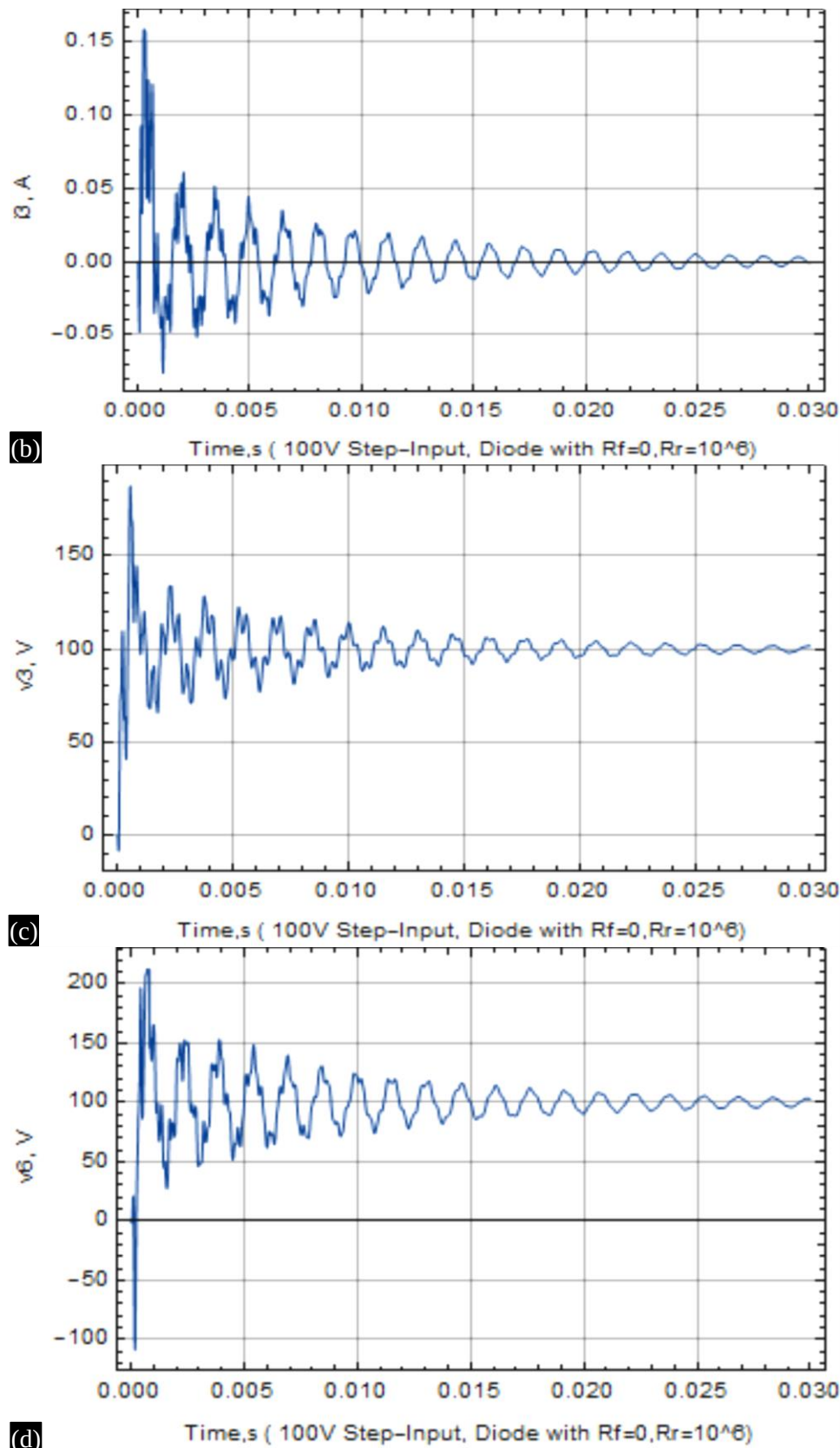


Figure 3. The response of a non-uniform winding excited by a 100-V step voltage source and terminated by a diode of $R_f=0$ and $R_r=1\text{M}\Omega$. (a) The source current, (b) the current i_3 , (c) the voltage v_3 , (d) the load voltage v_6 .

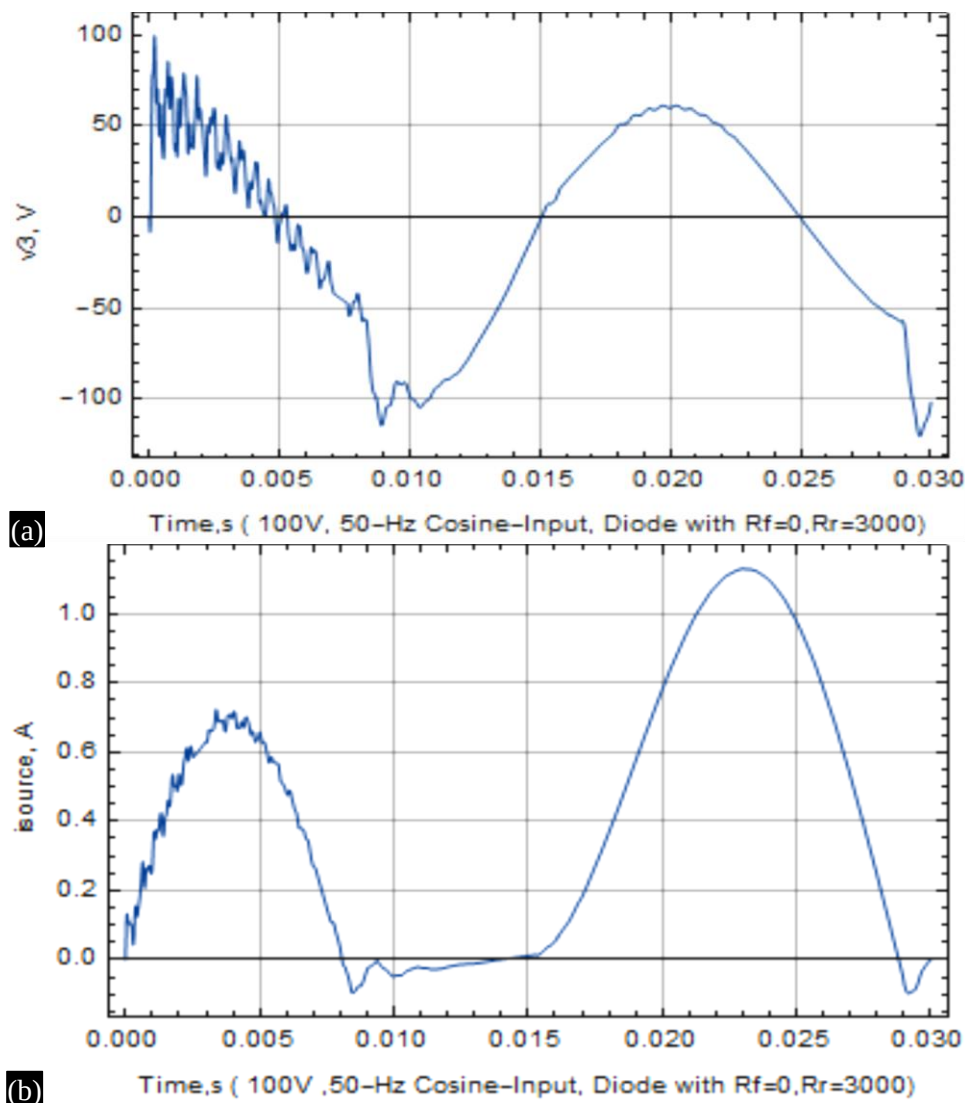
The Response of a Non-uniform Winding Excited By a 100-V 50-Hz Sinusoidal Voltage Source and Terminated by a Diode

This section deals with the transients of the non-uniform line due to a 50-Hz cosine function stimulus of a peak value 100V, i.e.

$$e(t) = 100 \cos 314t \quad (14)$$

Plot in Figure 4(a) illustrates the voltage v_3 at the junction between the second and third sections while the source current is depicted in plot in Figure 4(b). Both curves include initial high frequency transients superimposed on the main 50-Hz components. These fast transients decay rapidly and practically disappear after about 10 ms. The voltage v_3 oscillates between +80V and -130V.

The source current (i_{source}) and the load current (i_{load}) are shown in the two plots in Figure 4(b) and in Figure 4(c), respectively. They exhibit large currents in the forward direction due to the assumed zero forward resistance R_f and a relatively small current in the reverse direction because of the assumed very large (but finite) reverse resistance R_r . The load voltage v_6 , which is equal to the voltage across the diode, is shown in plot in Figure 4(d). The correspondence between the time plots of the diode current (Figure 4c) and its voltage (Figure 4d) is self-explanatory. The diode voltage is zero during the conduction periods, while the voltage assumes large negative reverse values during the zero current time intervals.



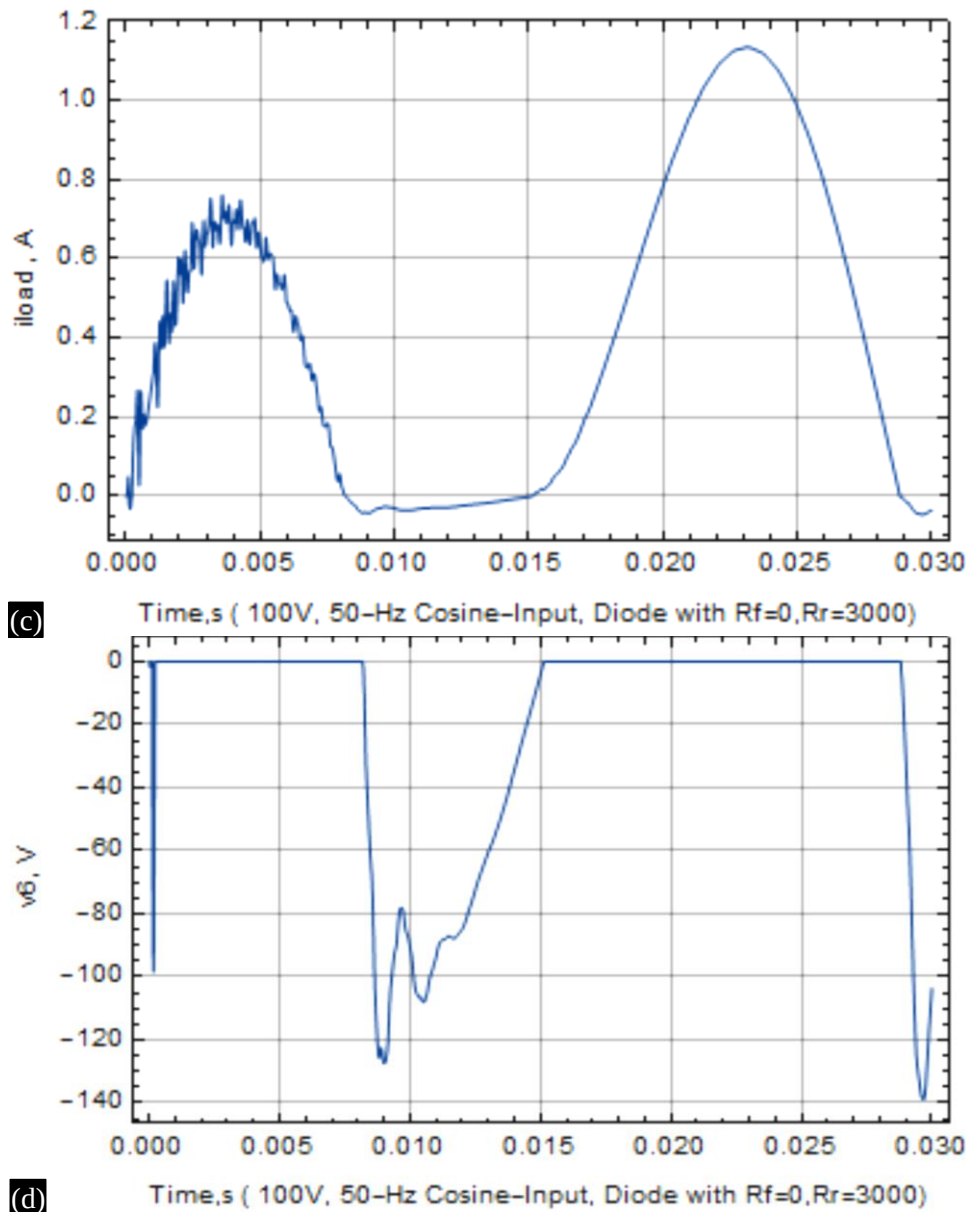


Figure 4. The response of a non-uniform winding excited by a 100-V 50-Hz sinusoidal voltage source and terminated by a diode of $R_f=0$ and $R_r=3k\Omega$. (a) The voltage v_3 , (b) the source current, (c) the load current, (d) the load voltage v_6 .

The Response of a Non-uniform Winding Excited By a 30-V Step Voltage Source and Terminated by a Non-linear Coil

In this case study, a 30-V step-voltage and the winding's capacitance values $c_g=8.5 \times 10^{-5}$ F and $c_s=3.4 \times 10^{-8}$ F were assumed.

The line's termination (at node 6) is a lossless non-linear inductor. Its characteristics are adopted from reference [1]. Its voltage-current relationship is assumed as follows:

$$V_6(t) = i_{load} \left[0.8 E^{\left(\frac{i_{load}}{1.76} \right)} \right] \quad (15)$$

Where E is the base of the natural logarithm (approximately 22/7) and the symbol $\wedge$ denotes an exponential operation. Equation (15) should replace equation (12) in the two previous case studies. The results are presented in Figure 5.

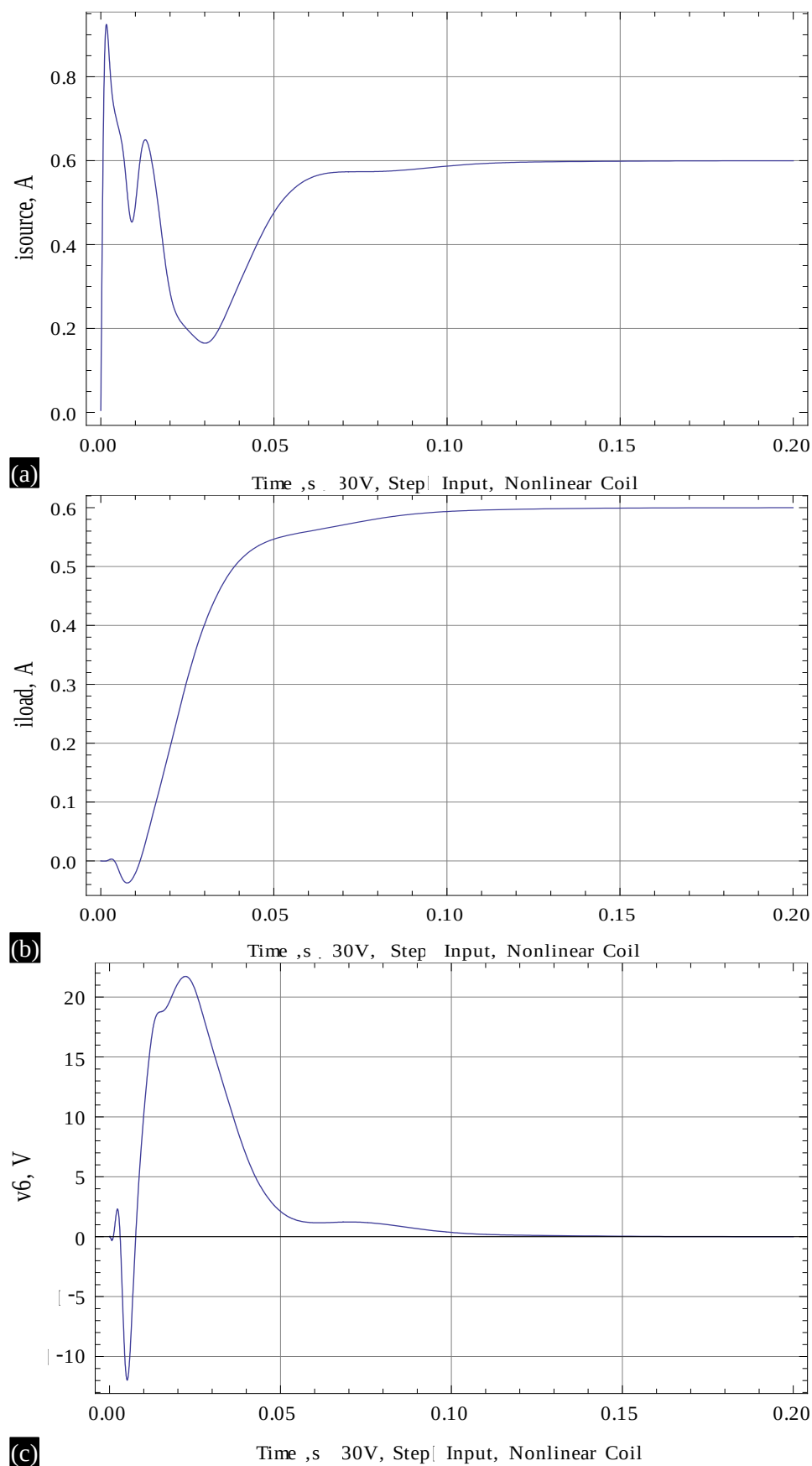


Figure 5. (a) The source current, (b) the load current, (c) the load voltage $v6$.

The source current is shown in plot in Figure 5(a). After initially oscillating between the two extreme values zero and 0.95A, it asymptotically approaches its final steady state magnitude 0.6A. This value agrees with the simple DC analysis of the 30-V source and the total winding resistance of 50 Ω . Plot in Figure 5(b) illustrates the termination current (i_{load}). As expected, it exhibits the same final value 0.6A. The current goes briefly in the negative range before the time point $t = 0.015$ s. The load voltage (v₆) is depicted in plot in Figure 5(c). It starts from the initial value zero and then oscillates between -12V and +22V before asymptotically approaching the expected final value zero.

CONCLUSIONS

The transients in a network comprising non-uniform transmission lines or machine windings terminated by non-linear loads are investigated in the time-domain.

The presented procedure can handle general non-uniformity and non-linearity situations and any time waveform of the sources initiating these transients. The results of several case studies are presented and discussed. They include the voltage and current distributions along the line or winding, the source and load currents in addition to the load voltage. Step-shaped as well as power frequency sinusoidal voltage stimuli are considered. As typical non-linear loads, ideal diode and a non-linear coil are analyzed. The investigation is conducted using the software *Mathematica*.

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